



APA analysis post processing

■ Date of Update	@August 14, 2025
■ Tags	
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📁 What Project are you updating?	 cmoore.3REAP
✓ What task are you updating?	DEXseq

Gene level summary

DEXSeq tests PAS (feature) usage within genes. To rank genes, collapse bin-level results:

- How many significant PAS per gene
- the strongest effect ($|LFC|$)
- direction agreement among significant PAS (all lengthen? mixed?)
- a single gene level FDR

Creating a collapsed results table:

- group the results table by groupID == gene ID
- padj cut off = 0.05
- LFC cut off = 0.5

Columns:

- `n_PAS` - number of PASs in group (gene)
- `n_sig` - number of PASs that pass the padj cut off (0.05)
- `n_big` - number of PASs that pass the LFC cutoff using absolute value (0.5)

- `min_padj` - the minimum padj value for the group
- `max_absL2FC` - the max absolute value LFC for the significant PASs in the group
- `mean_absL2FC` - the mean absolute value LFC for the significant PASs in the group
- `dir_consensus` - if all the significant PASs are one direction or mixed
- `perGeneQ` - a single gene-level FDR, calculated by DEXSeq
 - given per-exon p-values for null hypothesis H0, we can determine the number of genes in which at least for one exon H0 is rejected. What is the associated false discovery rate?

Notes about summary table:

- if a given gene has at least 1 significant PAS, then the rest of the values in the data table will be populated
- if a given gene has 0 significant PAS, then `max_absL2FC`, `mean_absL2FC`, and `dir_consensus` will be NA because those values are calculated for only significant PASs

Overall APA direction summary

```
> table(subset(res_3day, padj<=0.05)$APA_direction)
```

Lengthened	Shortened
113	105

```
> table(subset(res_5day, padj<=0.05)$APA_direction)
```

Lengthened	Shortened
134	94

Usage & average usage (per site, per condition)

DEXseq's fold change calculation is about usage, but it's very useful to show **observed usage**:

- per sample: `usage = PAS_count / sum(counts_allPAS_in_gene)`
- per condition: the mean usage across that condition's replicates
- overall "average usage" across all samples

Method:

- goal: convert raw counts into **usage** per PAS per sample, where usage is the fraction of a gene's reads that fall on a given PAS in that sample
- Step 1: do per-gene normalization to calculate PAS usage:
 - use groupID to sum counts across PAS within the same gene, per sample. This gives per-gene totals per sample.
 - broadcast each gene's per-sample total back to each PAS row belonging to that gene so each PAS row belonging to that gene has the correct denominator for that sample.
 - **create a usage matrix**: for each PAS and sample: `usage = PAS_count / sum(counts_allPAS_in_gene)`
 - rows as PAS, columns as samples
 - This ensures that **within a single sample, all PAS in a gene sum to 1**
- then, compute mean usage per PAS across samples
 - `meanUsage_All`
 - for each PAS, average its usage across all samples
 - summarizes overall PAS usage, not stratified by condition
 - `meanUsage_Control`
 - for each PAS, average its usage across samples in the control condition

- `meanUsage_Treatment`
 - for each PAS, average its usage across samples in the treatment condition
- in summary:
 - the values in each column (`meanUsage_*`) represents the **average fraction of reads assigned to that PAS, relative to all other PASs of its gene, in that condition grouping.**
 - they describe the **relative proportion (percent usage) of that PAS compared to other PASs in the same gene**, but at the *average sample level* rather than as a pooled total.
 - after averaging across samples (`meanUsage_*`) the PAS values per gene will be close to 1 but not guaranteed to sum exactly to 1 due to averaging across replicates and missing/unequal sample sets

Example: COL1A2 control sample usage

1. Counts for COL1A2

PAS	Rep1	Rep2	Rep3
PAS1	0	0	0
PAS2	0	0	0
PAS3	0	0	0
PAS4	0	0	0
PAS5	17	0	0

2. Per sample totals (denominator in usage calculation)

- Rep1 gene total = 17
- Rep2 gene total = 0 → forced to 1 to avoid div/0
- Rep3 gene total = 0 → forced to 1 to avoid div/0

3. Compute usage per sample

- This results in a usage matrix:

PAS	Rep1	Rep2	Rep3
PAS1	0/17 = 0	0/1 = 0	0/1 = 0
PAS2	0/17 = 0	0/1 = 0	0/1 = 0
PAS3	0/17 = 0	0/1 = 0	0/1 = 0
PAS4	0/17 = 0	0/1 = 0	0/1 = 0
PAS5	17/17 = 1	0/1 = 0	0/1 = 0

4. Average the usage across samples

PAS	Average usage
PAS1	mean(0,0,0) = 0
PAS2	mean(0,0,0) = 0
PAS3	mean(0,0,0) = 0
PAS4	mean(0,0,0) = 0
PAS5	mean(1,0,0) = 0.3333

Example: FUT4 control sample usage

1. Counts for FUT4

PAS	Rep1	Rep2	Rep3
PAS1	0	0	1
PAS2	0	0	0
PAS3	0	0	0
PAS4	16	8	14

2. Per sample totals (denominator in usage calculation)

- Rep1 gene total = 16
- Rep2 gene total = 8
- Rep3 gene total = 15

3. Compute usage per sample

- This results in a usage matrix:

PAS	Rep1	Rep2	Rep3
PAS1	0/16 = 0	0/8 = 0	1/15 = 0.06
PAS2	0/16 = 0	0/8 = 0	0/15 = 0
PAS3	0/16 = 0	0/8 = 0	0/15 = 0
PAS4	16/16 = 1	8/8 = 1	14/15 = 0.93

4. Average the usage across samples

PAS	Average usage
PAS1	mean(0,0,0.06) = 0.022
PAS2	mean(0,0,0) = 0
PAS3	mean(0,0,0) = 0
PAS4	mean(1,1,0.93) = 0.977

Plotting individual gene's relative PAS usage

- For each gene, get the mean usage for each condition (calculated above) and plot using bars
- Color by treatment
- Include PAS_ID and LFC on x-axis
 - LFC indicates change within that PAS between the conditions
- For each gene, there are multiple PASs, each has a genomic coordinate & strand. The distal PAS is the one farthest downstream in transcriptional direction, and that is indicated on the x-axis labels to quickly be able to see which PAS is "furthest" from the gene's transcription direction
 - on the + strand, that is the highest coordinate
 - on the - strand, that is the lowest coordinate